

Pricing American Options with LSMC: Replication and Extension

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1 1. Financial context and relevance

American options can be exercised at any time up to maturity, so pricing them is an **optimal stopping** problem under the risk-neutral measure. For vanilla American puts, early exercise can be rational (especially deep in-the-money and with positive rates). For high-dimensional payoffs (basket options, path dependence, multifactor models), lattice/PDE methods become expensive or intractable, making simulation-based methods essential.

The 2001 Longstaff-Schwartz paper introduced a practical simulation approach (Least-Squares Monte Carlo, LSMC) that estimates continuation values via cross-sectional regressions and then applies backward dynamic programming. This became one of the most influential methods in buy-side/sell-side derivatives libraries.

2 2. Primary and extension papers

2.1 Primary paper

- Longstaff, F. A., & Schwartz, E. S. (2001). *Valuing American Options by Simulation: A Simple Least-Squares Approach*. The Review of Financial Studies.
- URL: <https://galton.uchicago.edu/~mykland/346W07/Longstaff.pdf>

2.2 Related/extended paper

- Tsitsiklis, J. N., & Van Roy, B. (2001). *Regression Methods for Pricing Complex American-Style Options*. IEEE Transactions on Neural Networks.
- URL: <https://www.mit.edu/~jnt/Papers/J086-01-bvr-options.pdf>

3 3. Methodology implemented

3.1 Longstaff-Schwartz (LSMC)

American options are more difficult to price than European options because the holder may exercise before maturity. This early-exercise feature turns valuation into an optimal stopping problem.

The Longstaff-Schwartz least-squares Monte Carlo (LSMC) method combines Monte Carlo simulation with regression in order to approximate the continuation value. The key idea is

that the continuation value is not directly observable during simulation, so it is estimated statistically from simulated future cash flows.

For an American put, the immediate exercise value at time t is the intrinsic value.

$$(K - S_t)^+$$

The continuation value can be written as the conditional expectation of discounted future payoffs.

$$C(S_t) = \mathbb{E} [e^{-r\Delta t} V_{t+1} \mid S_t]$$

LSMC approximates this conditional expectation by regressing discounted realized future cash flows on basis functions of the current state variable.

The algorithm starts by simulating a large number of stock-price paths under the risk-neutral dynamics. At maturity, the option value is simply the terminal payoff.

A practical feature of the Longstaff-Schwartz method is that the regression is usually restricted to in-the-money paths. The reason is that when the option is out of the money, immediate exercise is irrelevant, so those paths contribute little useful information about the stopping decision.

In our implementation, we apply LSMC to price an American put option under a geometric Brownian motion model. The initial stock price is based on recent AAPL data, the volatility is estimated from five years of historical log returns, and the short rate is approximated using the $\sim$ IRX yield as a simple proxy.

3.1.1 Practical implementation choices

Several practical implementation choices affect the behavior of the LSMC method. First, the regression basis must be rich enough to capture the shape of the continuation value, but not so flexible that it introduces instability or overfitting. In our implementation, we use Laguerre-type basis functions, which are a standard choice in the literature because they provide a simple polynomial approximation while remaining computationally convenient.

Second, the regression is restricted to in-the-money paths. This choice is motivated by the fact that the stopping decision is only relevant when immediate exercise has positive value. Excluding out-of-the-money paths reduces noise in the regression and concentrates the approximation on the region that matters most for exercise. However, this also means that the quality of the fitted continuation value depends on the number and distribution of in-the-money observations at each time step.

Third, the number of simulated paths and the number of exercise dates both influence numerical stability. More paths reduce Monte Carlo noise, while a finer time discretization produces a closer approximation to continuous-time early exercise. In practice, these choices create a trade-off between computational cost and pricing accuracy.

3.1.2 Method limitations and approximation risk

Although LSMC is flexible and widely used, it is still an approximation method and therefore subject to several sources of error. One important source is basis risk: if the chosen regression basis does not represent the continuation value well, then the stopping rule may be distorted and the option price may be biased. Another source is simulation noise, since the regression is built on realized future cash flows rather than exact conditional expectations.

The method may also become unstable in regions where the number of in-the-money paths is small. This issue is particularly relevant for early exercise dates or contracts that are far out of the money, where the regression sample may be limited. In addition, because the stopping rule is estimated recursively backward through time, small approximation errors at one step can affect later decisions and propagate through the pricing procedure. These limitations do not invalidate the method, but they help explain why implementation choices such as basis design, path count, and discretization matter so much in practice.

3.2 Tsitsiklis-Van Roy style regression DP (extension)

The Tsitsiklis-Van Roy approach studies American-style option pricing from the perspective of approximate dynamic programming. Instead of focusing primarily on the stopping decision through in-the-money regression, this framework emphasizes the recursive approximation of the option value over time. The motivation is that exact dynamic programming becomes computationally infeasible when the state space is large, especially for derivatives with multiple sources of uncertainty or more complex payoff structures.

The central idea is to approximate the value function using a finite set of basis functions and then update this approximation recursively through simulated paths. In this sense, the method is closely related to fitted value iteration. At each time step, the discounted continuation value from the next step is projected onto a chosen function space, and the fitted function is used as the approximation of the current value function. This provides a dynamic-programming interpretation of regression-based pricing and highlights the connection between simulation and value-function approximation.

Compared with Longstaff-Schwartz, the Tsitsiklis-Van Roy style method is more global in its approximation structure. Longstaff-Schwartz typically performs regression only on in-the-money paths because those paths are directly relevant for the exercise decision. By contrast, the Tsitsiklis-Van Roy framework is often implemented using regression over all simulated paths, which makes the approximation less tied to a local stopping region and more closely

aligned with recursive value-function estimation. This provides a useful comparison benchmark, although it may also make the results more sensitive to basis specification and approximation error.

An important contribution of the Tsitsiklis-Van Roy paper is its theoretical discussion of error behavior. The paper shows that approximation error can accumulate severely if representative states are chosen poorly. On the other hand, when simulated trajectories follow the underlying risk-neutral state dynamics, the approximation becomes more stable. This gives the method stronger theoretical support and helps explain why simulation-based regression methods can remain useful even when exact dynamic programming is infeasible.

In our project, the Tsitsiklis-Van Roy style comparator is implemented as a fitted-value-iteration benchmark alongside LSMC. We regress discounted next-step continuation values on polynomial basis functions using all simulated paths, and we use the resulting fitted function as an alternative estimate of the option value. This allows us to compare a more classical exercise-focused regression method with a more global dynamic-programming-based approximation, and it helps us interpret the pricing differences observed in the numerical results.

3.2.1 Relation to approximate value iteration

A useful way to understand the Tsitsiklis-Van Roy framework is to view it as a finite-horizon version of approximate value iteration. Instead of attempting to compute the exact value function at every possible state, which is generally infeasible in higher dimensions, the method projects the value function onto a lower-dimensional space generated by basis functions. The resulting fitted function is then used in place of the exact continuation value when stepping backward through time.

This perspective is important because it emphasizes that the method is not merely a numerical shortcut, but a structured approximation to dynamic programming. In other words, the goal is not only to estimate whether exercise is optimal at a particular point, but to approximate the value of continuation throughout the relevant state space. This makes the Tsitsiklis-Van Roy approach conceptually broader than a local regression rule tied only to the immediate stopping decision.

3.2.2 Approximation error and methodological trade-offs

The theoretical importance of the Tsitsiklis-Van Roy paper lies partly in its treatment of approximation error. The paper shows that when value-function approximation is performed carelessly, errors can accumulate over time and may even grow rapidly with the horizon. This is especially relevant in finite-horizon American-style option pricing, where recursive approximation is repeated over many exercise dates.

At the same time, the paper argues that if representative states are generated from simulated risk-neutral trajectories, the approximation becomes more stable. This insight provides a theoretical justification for simulation-based regression methods and explains why they can remain useful despite the curse of dimensionality. In practical terms, however, the method still faces trade-offs. A more global approximation can provide smoother value estimates, but it may also become more sensitive to basis selection and model misspecification. This is one reason why comparing the Tsitsiklis–Van Roy style method with LSMC is useful in our project: the comparison highlights the difference between a more exercise-focused local regression procedure and a more global recursive value-function approximation.

4 4. Data and experimental setup

Our empirical study uses Apple Inc. (AAPL) as the underlying asset. Historical price data are obtained from Yahoo Finance through the `yfinance` package. We use approximately five years of daily closing prices in order to estimate the current spot level and historical volatility. The use of a large-cap and highly traded stock makes the example easy to interpret and provides a realistic market context for illustrating American-style option pricing methods.

The current stock price is taken as the most recent closing price in the downloaded sample. Volatility is estimated from the standard deviation of daily log returns and then annualized using the square-root-of-time rule. Although this historical estimate is simple and does not reflect the full implied-volatility surface used in practice, it provides a transparent and reproducible input for a research-oriented implementation. As a proxy for the risk-free rate, we use the most recent value of the 13-week U.S. Treasury bill index ($\text{\textasciitilde TRX}$), scaled into decimal form. This choice is consistent with the simplified risk-neutral setting used in our implementation.

To generate numerical option values, we assume that the underlying stock follows a geometric Brownian motion under the risk-neutral measure with constant volatility and constant interest rate. This assumption is standard in introductory Monte Carlo option-pricing exercises and makes it possible to compare the pricing behavior of LSMC and the Tsitsiklis–Van Roy style comparator in a controlled setting. At the same time, we recognize that this assumption is restrictive, since real equity dynamics may exhibit time-varying volatility, jumps, and other deviations from the Black–Scholes framework.

In our experiments, the option maturity is set to one year and the time horizon is divided into 50 exercise dates. This discretization gives a reasonable Bermudan-style approximation to a one-year American option while keeping the computation manageable. We simulate 100,000 stock-price paths in order to reduce Monte Carlo noise and obtain relatively stable price estimates across methods.

To examine pricing behavior across different moneyness levels, we consider three strike prices chosen around the current AAPL spot price. This allows us to compare how both methods behave for relatively lower-strike, near-the-money, and higher-strike American put contracts.

Using several strikes rather than only one contract makes the comparison more informative, since the quality of continuation-value approximation may vary across payoff regions.

For benchmarking purposes, we also compute the corresponding European put prices under the Black–Scholes model using the same spot price, volatility, maturity, strike, and risk-free rate inputs. These European prices do not represent directly comparable contracts in terms of exercise flexibility, but they provide a useful lower reference point, since an American put should not be worth less than its European counterpart under the same model assumptions. This benchmark also serves as a basic numerical consistency check before comparing the two regression-based American pricing methods.

Overall, this experimental design is intended to balance realism, interpretability, and reproducibility. The market inputs are based on observable AAPL data, while the numerical environment remains simple enough to highlight the differences between the two regression-based pricing methods.

5 5. Python implementation and project structure

- `data/` : raw/derived data (`aapl_prices.csv`, `pricing_results.csv`)
- `src/` : implementation scripts (`lsmc.py`, `tsitsiklis_vanroy.py`, `data_loader.py`, `run_study.py`)
- `visuals/` : generated plots
- `report/` : this QMD report

6 6. Results

```
import pandas as pd
results = pd.read_csv('../data/pricing_results.csv')
results
```

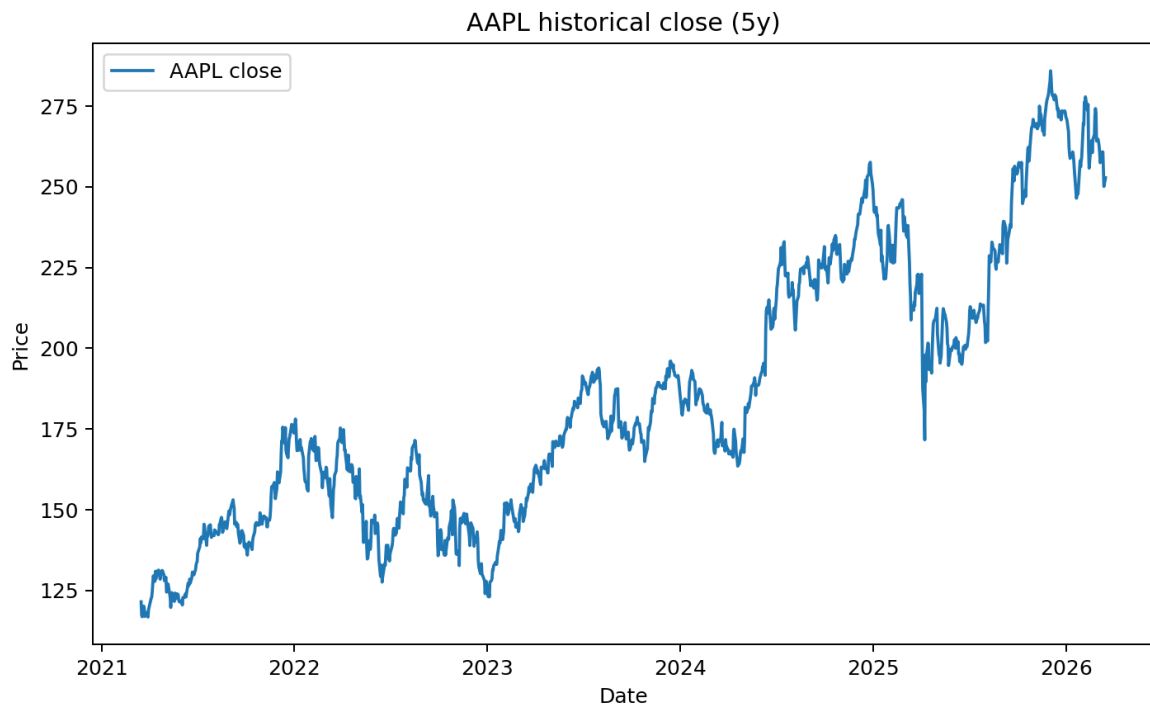
	ticker	spot	strike	maturity_years	volatility	risk_free	european_put_bs	lsmc_pric
0	AAPL	252.820007	227.538007	1.0	0.274248	0.03605	12.409216	12.712565
1	AAPL	252.820007	252.820007	1.0	0.274248	0.03605	22.842482	23.563214
2	AAPL	252.820007	278.102008	1.0	0.274248	0.03605	36.790064	38.313742

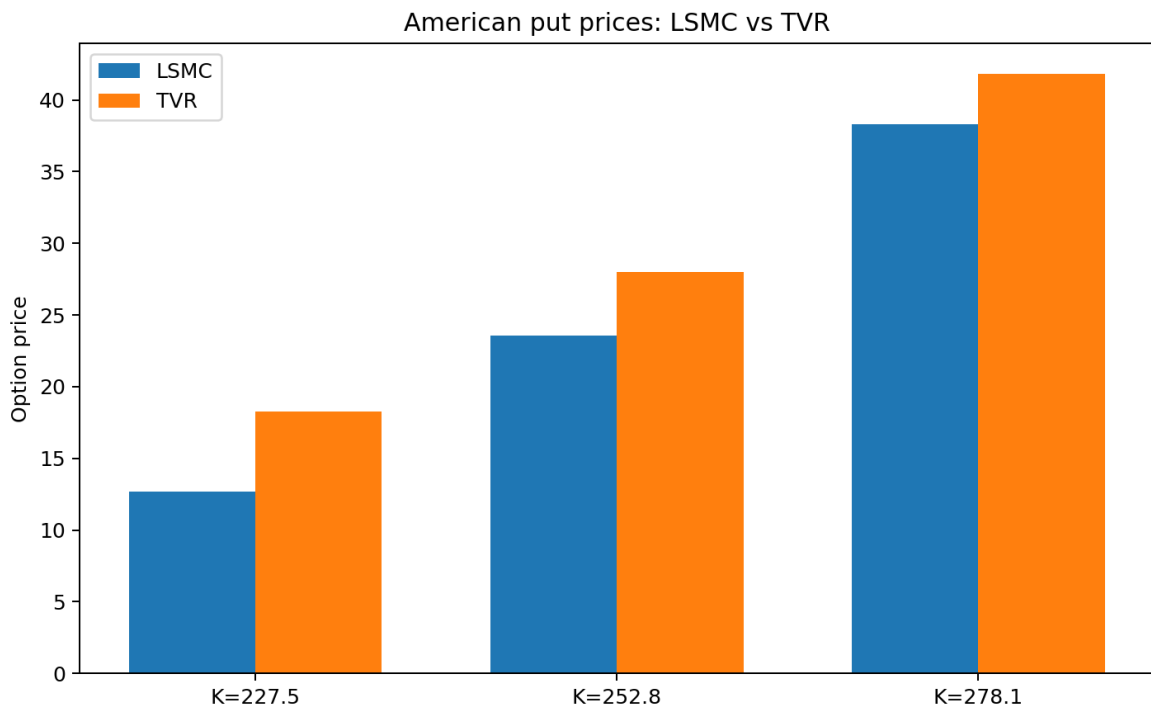
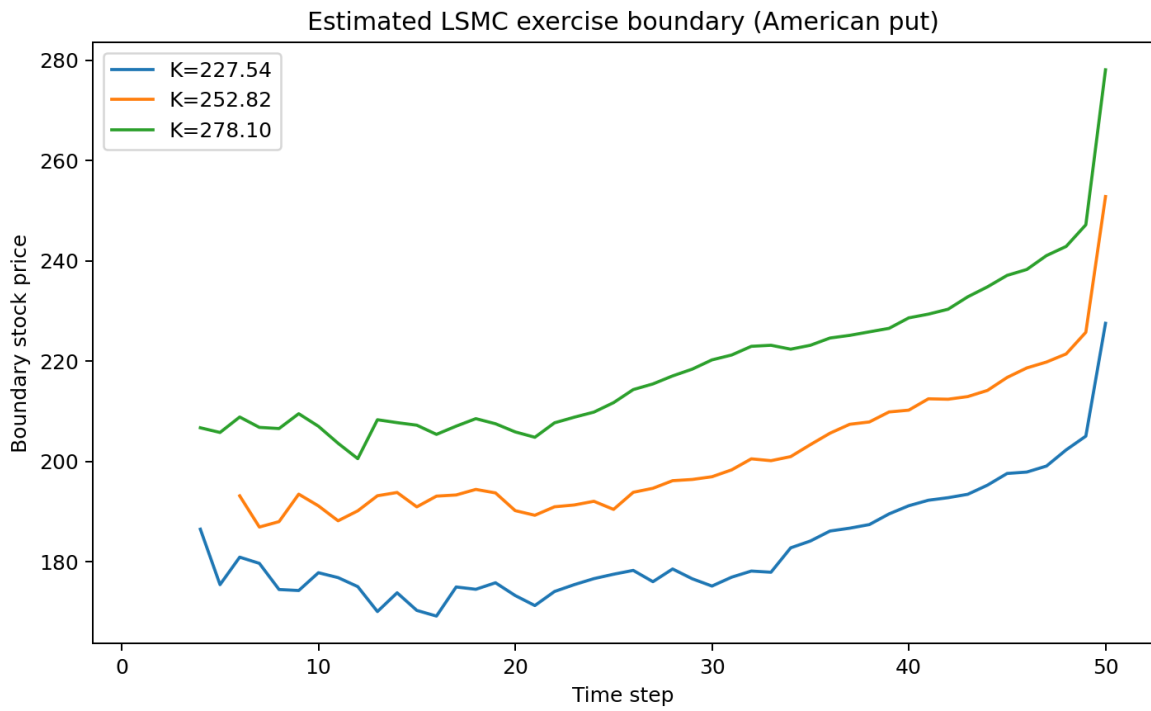
A quick consistency check is that American put values should be at least as large as corresponding European put values (same model inputs); this holds in the generated output.

```
(results[['strike','european_put_bs','lsmc_price','tvr_price']])
```

	strike	european_put_bs	lsmc_price	tvr_price
0	227.538007	12.409216	12.712565	18.257806
1	252.820007	22.842482	23.563214	27.985602
2	278.102008	36.790064	38.313742	41.849501

6.1 Visuals





7 7. Critical evaluation

7.1 Strengths of LSMC / regression simulation methods

- Scales better than trees/PDE in high dimensions.
- Flexible: supports path-dependent state variables and custom payoff logic.
- Natural fit for enterprise Monte Carlo risk stacks.
- Easy to parallelize.

7.2 Weaknesses / limitations

- **Basis risk / approximation error:** poor basis choice biases continuation estimates.
- **Look-ahead and finite-sample bias** risk if implementation is careless.
- **Exercise boundary instability** in sparse ITM regions.
- **Model risk:** under realistic markets, GBM with constant volatility/rates is often insufficient.
- **Tail events** underrepresented in standard MC unless variance reduction / stress design is used.

7.3 Does this address real-world needs appropriately?

Partly yes. LSMC is practically valuable and remains widely used. But institutions using it naively (simple GBM, fixed polynomial basis, weak diagnostics) may systematically misprice early-exercise features and risk exposures, especially in stressed regimes or products with complex state dependence.

7.4 Where institutions can run into trouble

- Underfitting continuation values for deep OTM/ITM states.
- Overfitting with high-order basis and limited effective sample.
- Unstable Greeks due to regression noise.
- Governance gaps: insufficient backtesting against benchmarks and market quotes.

7.5 Suggestions for improvement

1. **Richer dynamics:** local/stochastic volatility, stochastic rates, jumps when needed.
2. **Regularized / adaptive regression:** orthogonalized basis, cross-validation, or modern ML regressors with strong controls.
3. **Variance reduction:** antithetic/control variates, quasi-MC.

4. **Dual bounds** (e.g., Andersen-Broadie style) to quantify pricing bias intervals.
5. **Robust model validation**: benchmark against lattice/PDE in low dimensions, scenario stress testing, and out-of-sample stability checks.

8 8. Reproducibility

From project root:

```
python -m pip install yfinance pandas numpy matplotlib
python src/run_study.py
quarto render report/american_options_lsmc_study.qmd --to pdf
```

If PDF rendering tools are unavailable, render to HTML as fallback:

```
quarto render report/american_options_lsmc_study.qmd --to html
```